MPGV Monitor — System Overview

Dynamic Visualization of Seismic Data for Iceland

Links

• Local Dev (Vite): http://localhost:5173
• Local API: http://localhost:5000
• Production (Pluto): http://pluto.cs.hi.is:9001

Purpose

MPGV Monitor continuously collects earthquakes around Iceland (Mw ≥ 3.0), persists them in SQLite, and renders an interactive map to explore temporal and spatial patterns by time window and magnitude, with optional volcano overlays.

Architecture at a Glance

Data Flow

1) **scrape.py** retrieves recent year/month pages from the Icelandic Met Office (Hraun Vedur), parses tables with BeautifulSoup, normalizes timestamps, filters Mw<3.0, and attempts inserts. 2) A **UNIQUE(date_time, latitude, longitude)** key (plus INSERT OR IGNORE) prevents duplicates. 3) **APScheduler** runs the scrape periodically (every few minutes), appending only new events. 4) The frontend queries **/earthquakes**, **/volcanoes**, and **/deep-earthquakes** to render the map.

API Endpoints

GET /earthquakes	All quakes; newest first (JSON array).
GET /deep-earthquakes	Quakes with depth ≥ 25 km; sorted by depth desc.
GET /volcanoes	Volcano catalog with coordinates and metadata.
POST /update-earthquakes	Manual scrape trigger (optional in dev).
GET /scrape-volcanoes	Maintenance endpoint to (re)populate volcano data.

Frontend Components & Behavior

TimeWindowSlider: zoomable timeline (years → months → weeks → days), drag to pan; thick year dividers and mid-year labels for readability.
MagnitudeScale: vertical slider to filter minimum Mw; value chip follows the thumb.
MapComponent: Google Maps rendering of quakes (color by magnitude), volcano toggles; higher-magnitude markers are drawn on top for clarity.

Database Schema (SQLite)

Table	Columns (key fields bold)
earthquake	date_time (TEXT), latitude (REAL), longitude (REAL), depth (REAL), m
volcano	name (TEXT), description (TEXT), elevation_m/ft (REAL/INT), latitude (REAL),

Operational Health & Notes

• No exact duplicates by design; occasional same-timestamp near-duplicates may occur from the source feed. • Scheduler cadence decouples scraping from user reads; DB grows incrementally. • In production, frontend and API are same-origin (no CORS hop).

Verification (Pluto)

```
curl -I http://pluto.cs.hi.is:9001  (200 OK)
curl -s http://pluto.cs.hi.is:9001/earthquakes | head
sqlite3 backend/data/earthquakes.db 'SELECT COUNT(*), MIN(date_time), MAX(date_time) FROM eart
```

This document summarizes the current architecture and behavior of MPGV Monitor in both development and production deployments.